

Block-diagonal block-encoding vs K (fixed N = 256): Clader-Frobenius vs SVD

Task: block-encode $A_{\text{block}} = \sum_k |k\rangle\langle k| (x) A_k$, each A_k of size $N \times N$ with $N = 256 = 2^8$, sweeping the number of blocks K in $[1, 2, 4, 8, 16, 27, 32, 64, 125, 128, 216]$ ($1 \dots 216 = 6^3$), $\text{phase_bitsize} = 32$. Two methods, each Toffoli-optimal and qubit-optimal.

- (1) CLADER-FROBENIUS BLOCK ($\alpha = F_{\text{max}} = \max_k ||A_k||_F$): a read-only k register addresses every QROAM; a k -controlled flag rotation injects F_k/F_{max} , then the k -controlled Frobenius construction $U_R^{\dagger} U_L$ prepares the row/column states.
T-opt : Toffoli-minimal QROAM select-swap batch (more ancilla).
Q-opt : un-batched $\lambda = 1$ QROAM (fewest qubits, more T).
- (2) SVD INTERFEROMETER BLOCK ($\alpha = 1$): $A_k = U_k \Sigma_k V_k$ via the block-unitary interferometer + QROAM diagonal; each metric optimized over a QROAM batching λ (blocking the K -block dimension).

SCALING FITS $y = c * K^a$ (slope a per doubling of K ; N fixed):

Clader-Frobenius block, Toffoli-opt	T: $a=0.50$	Q: $a=0.51$
Clader-Frobenius block, qubit-opt	T: $a=1.00$	Q: $a=0.03$
SVD interferometer block, Toffoli-opt	T: $a=0.45$	Q: $a=0.39$
SVD interferometer block, qubit-opt	T: $a=0.90$	Q: $a=0.02$

Reading the K -dependence (the fits above bear this out):

- * Toffoli-OPT: $T \sim \sqrt{K}$ ($a \sim 0.5$ for both methods). The select-swap QROAM amortizes the $K \times N \times N$ data load across blocks -- its optimum batch $\lambda_{\text{batch}} \sim \sqrt{K N / b}$ gives cost $\sim \sqrt{K N b}$, so doubling K only raises T by $\sim \sqrt{2}$. This block-amortization is the whole point of select-swap.
- * Qubit-OPT ($\lambda = 1$): $T \sim K$ ($a \sim 1.0$; ~ 0.9 for SVD). With no batching every block's data is loaded serially, so Toffoli is linear in K -- but qubit count stays $\sim \text{flat}$ in K ($a \sim 0.02-0.03$): only $\log_2(K)$ address qubits are added.
- * Clader T-opt qubit count grows with K ($a \sim 0.44$): its select-swap batch widens with the $K \times N \times N$ data. The k register itself adds only $\log_2(K)$ qubits.

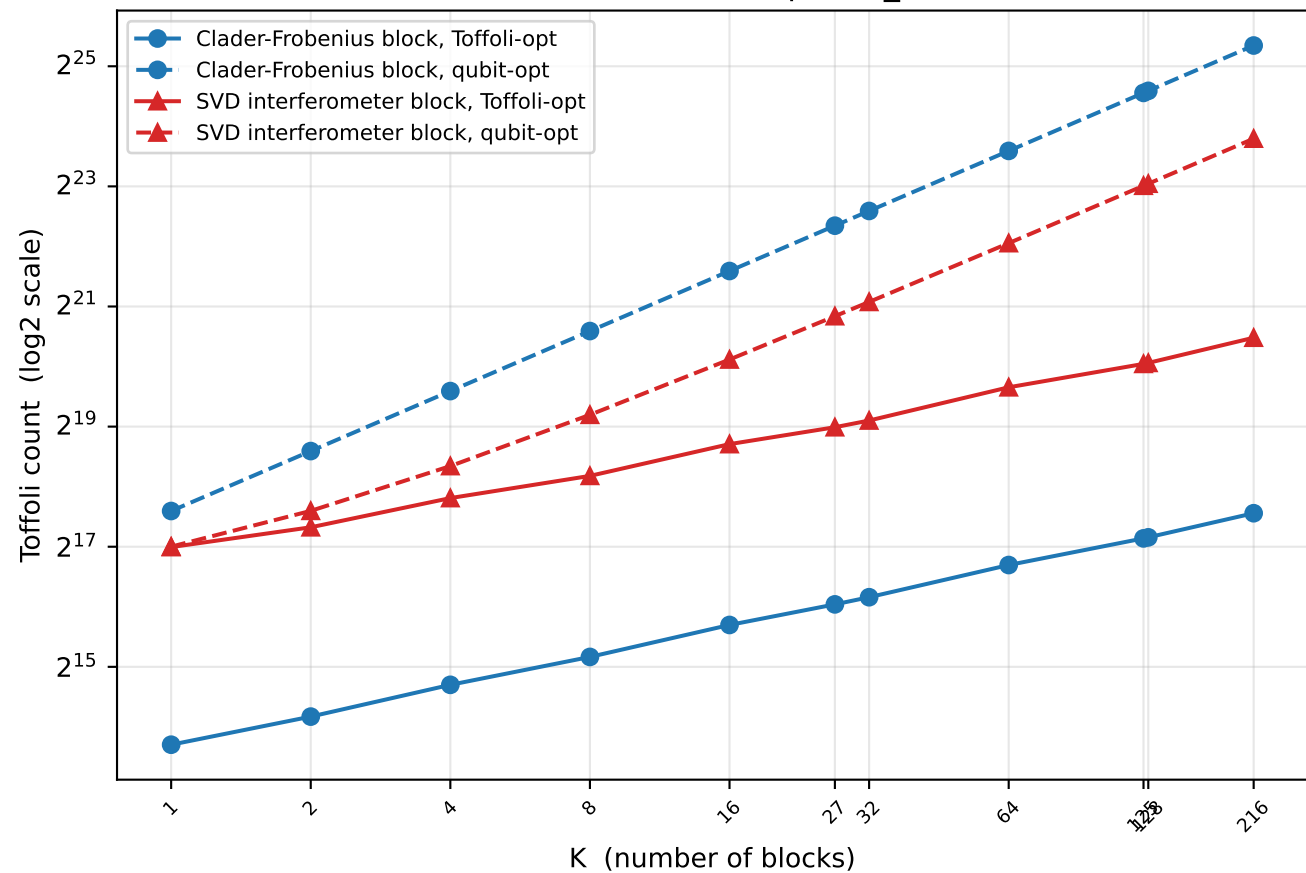
KEY CAVEAT - subnormalization is NOT equal: Clader-block $\alpha = F_{\text{max}}$ ($\sim N$, independent of K) vs SVD-block $\alpha = 1$. These are per-query costs; downstream algorithms need $\sim \alpha$ queries, so amortize the Clader cost by F_{max} .

Counts from Qualtran $\text{QECGatesCost} / \text{QubitCount}$; error $\sim O(2^{-\text{phase_bitsize}})$.

Toffoli (T) / qubit (Q) counts vs K: $\sum_k |k\rangle\langle k|$ (x) A_k, N = 256
 phase_bitsize = 32; alpha(Clader) = F_max, alpha(SVD) = 1

K	Clader-Frobenius Toffoli-opt	Clader-Frobenius qubit-opt	SVD interferometer Toffoli-opt	SVD interferometer qubit-opt
1	T=13,339 Q=1,085	T=197,938 Q=98	T=130,508 Q=174	T=131,177 Q=111
2	T=18,449 Q=2,110	T=395,316 Q=100	T=164,044 Q=176	T=198,249 Q=113
4	T=26,623 Q=2,112	T=790,074 Q=102	T=229,778 Q=305	T=332,393 Q=115
8	T=36,739 Q=4,161	T=1,579,590 Q=104	T=296,850 Q=307	T=600,681 Q=117
16	T=53,131 Q=4,163	T=3,158,622 Q=106	T=428,318 Q=564	T=1,137,257 Q=119
27	T=67,383 Q=8,260	T=5,329,791 Q=108	T=520,542 Q=566	T=1,875,049 Q=121
32	T=73,188 Q=8,260	T=6,316,686 Q=108	T=562,462 Q=566	T=2,210,409 Q=121
64	T=106,116 Q=8,262	T=12,632,814 Q=110	T=825,398 Q=1,079	T=4,356,713 Q=123
125	T=144,331 Q=16,455	T=24,672,933 Q=112	T=1,081,110 Q=1,081	T=8,448,105 Q=125
128	T=146,086 Q=16,455	T=25,265,070 Q=112	T=1,093,686 Q=1,081	T=8,649,321 Q=125
216	T=192,945 Q=16,457	T=42,634,422 Q=114	T=1,462,582 Q=1,083	T=14,551,657 Q=127

Toffoli count vs K (N = 256, phase_bitsize = 32)



Qubit count vs K (N = 256, phase_bitsize = 32)

